
EECS 16A
Fall 2025

Foundations of Signals, Systems, & Information Processing

Discussion 3A

1. Signal Fundamental Frequency

For each of the discrete-time signals below, determine the fundamental period (and corresponding fundamental frequency), or explain why the signal is not periodic.

- (a) $x[n] = e^{i\frac{\pi}{5}n}$ for all integers n .

Answer: We want to find the smallest positive integer p such that $x[n+p] = x[n]$ for all integers n .

$$x[n+p] = e^{i\frac{\pi}{5}(n+p)} = e^{i\frac{\pi}{5}n} e^{i\frac{\pi}{5}p} = e^{i\frac{\pi}{5}n} = x[n].$$

Canceling the common exponentials involving n from each side, we have

$$e^{i\frac{\pi}{5}p} = 1.$$

Therefore, we must have

$$\frac{\pi}{5}p = 2\pi\ell$$

where ℓ is the smallest positive integer ℓ that leads to a positive integer value for p . We have

$$p = 10\ell,$$

so we choose $\ell = 1$, which leads to the fundamental period

$$p = 10$$

and corresponding fundamental frequency

$$\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{10} = \frac{\pi}{5}.$$

- (b) $x[n] = \cos\left(\frac{3\pi}{5}n\right)$ for all integers n .

Answer: We want to find the smallest positive integer p such that $x[n+p] = x[n]$ for all integers n .

$$\begin{aligned} x[n+p] &= \cos\left(\frac{3\pi}{5}(n+p)\right) \\ &= \cos\left(\frac{3\pi}{5}n + \frac{3\pi}{5}p\right) = \cos\left(\frac{3\pi}{5}n\right) = x[n]. \end{aligned}$$

Therefore, we must have

$$\frac{3\pi}{5}p = 2\pi\ell$$

where we seek the smallest positive integer ℓ that leads to a positive integer value for p . We have

$$p = \frac{10\ell}{3},$$

so we choose $\ell = 3$, which leads to the fundamental period

$$p = 10$$

and corresponding fundamental frequency

$$\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{10} = \frac{\pi}{5}.$$

- (c) $x[n] = \sin(n)$ for all integers n .

Answer: This sinusoid is *not* periodic, because no integer p can be found such that $x[n+p] = x[n]$ for all integers n . For a sinusoid x described by, say, $x[n] = \sin(\omega_x n)$ to be periodic, ω_x must be a rational multiple of π . In our case, $\omega_x = 1$, which is not a rational multiple of π . So, x is not periodic.

This can be seen as follows. We want to find the smallest positive integer p such that $x[n+p] = x[n]$ for all integers n .

$$x[n+p] = \sin\left(\omega_x(n+p)\right) = \sin\left(\omega_x n + \omega_x p\right) = \sin(\omega_x n) = x[n].$$

Therefore, we must have

$$\omega_x p = 2\pi\ell$$

where we seek the smallest positive integer ℓ that leads to a positive integer value for p . We have

$$p = \frac{2\pi}{\omega_x} \ell.$$

Since $\omega_x = 1$, we have

$$p = 2\pi\ell.$$

Clearly, no integer value of ℓ exists that can make p an integer. So, the signal is not periodic. For the signal to be periodic, we must have ω_x to be a rational multiple of π , so it cancels the π in the numerator. In other words, if

$$\omega_x = \frac{M}{N}\pi,$$

where M and N are coprime (have no factors in common), then

$$p = \frac{2\pi}{\omega_x} \ell = \frac{2\pi}{\frac{M}{N}\pi} \ell = \frac{2N}{M} \ell.$$

We then select the smallest positive integer ℓ that makes p an integer.

2. Fear Not the Complex Exponential!

Consider a signal x where $x[n] = e^{i\frac{2\pi}{5}n} + 2e^{-i\frac{4\pi}{5}n} + 3e^{i\frac{6\pi}{5}n}$, $\forall n \in \mathbb{Z}$

(a) Determine whether x is periodic. If it is periodic, find the fundamental period p .

Answer: Each term in $x[n]$ is a complex exponential of the form $e^{i\omega n}$, where ω is a rational multiple of 2π . A DT signal of this form is periodic if each frequency component is rational with respect to 2π . Let's first try finding the period of each individual term:

$$e^{i\frac{2\pi}{5}n} \rightarrow \text{Fundamental period } p_1 = \frac{2\pi}{\frac{2\pi}{5}} = 5$$

$$e^{-i\frac{4\pi}{5}n} \rightarrow p_2 = \frac{2\pi}{\frac{4\pi}{5}} = \frac{2\pi}{\frac{4\pi}{5}} = \frac{2\pi \cdot 5}{4\pi} = \frac{10}{4} = 2.5 \text{ (Not integer)}$$

$$\text{But: } e^{-i\frac{4\pi}{5}n} = e^{i\frac{6\pi}{5}n} \text{ (since } e^{-i\theta} = e^{i(2\pi-\theta)}) \Rightarrow \text{Same as third term.}$$

Another way to think about this equality can be found in the solution for part (c)

For $e^{i\frac{6\pi}{5}n}$:

$$\omega = \frac{6\pi}{5} \Rightarrow \frac{6\pi}{5} = \frac{2\pi \cdot 3}{5} \Rightarrow p_3 = \frac{2\pi}{\frac{6\pi}{5}} = \frac{10}{6} = \frac{5}{3} \text{ (Not integer).}$$

We instead look for a common integer p such that all terms complete full cycles:

$$\omega_1 = \frac{2\pi}{5}, \quad \omega_2 = -\frac{4\pi}{5}, \quad \omega_3 = \frac{6\pi}{5}$$

All these frequencies are rational multiples of 2π , so $x[n]$ is periodic. Let's express each frequency as a rational number times 2π :

$$\omega_1 = \frac{2\pi}{5} = 2\pi \cdot \frac{1}{5}, \quad \omega_2 = -\frac{4\pi}{5} = 2\pi \cdot \left(-\frac{2}{5}\right), \quad \omega_3 = \frac{6\pi}{5} = 2\pi \cdot \frac{3}{5}$$

The least common multiple of the denominators (5) gives the fundamental period:

$$p = 5$$

So, $x[n]$ is periodic with fundamental period $p = 5$.

- (b) Simplify x using Euler's formula and express your answer in terms of real sinusoids in the form $A + iB$, where A is the real part and B is the imaginary part.

Hint: Euler's formula is given by $e^{i\theta} = \cos \theta + i \sin \theta$

Answer: Using Euler's formula, we rewrite each term:

$$\begin{aligned} e^{i\frac{2\pi}{5}n} &= \cos\left(\frac{2\pi}{5}n\right) + i \sin\left(\frac{2\pi}{5}n\right) \\ 2e^{-i\frac{4\pi}{5}n} &= 2 \left[\cos\left(\frac{4\pi}{5}n\right) - i \sin\left(\frac{4\pi}{5}n\right) \right] \\ 3e^{i\frac{6\pi}{5}n} &= 3 \left[\cos\left(\frac{6\pi}{5}n\right) + i \sin\left(\frac{6\pi}{5}n\right) \right] \end{aligned}$$

Grouping real and imaginary parts:

$$\begin{aligned} x[n] &= \cos\left(\frac{2\pi}{5}n\right) + 2\cos\left(\frac{4\pi}{5}n\right) + 3\cos\left(\frac{6\pi}{5}n\right) \\ &\quad + i \left[\sin\left(\frac{2\pi}{5}n\right) - 2\sin\left(\frac{4\pi}{5}n\right) + 3\sin\left(\frac{6\pi}{5}n\right) \right] \end{aligned}$$

So, the simplified form is:

$$\begin{aligned} x[n] &= \text{Re}\{x[n]\} + i \text{Im}\{x[n]\}, \text{ with the above expressions} \\ A = \text{Re}\{x[n]\} &= \cos\left(\frac{2\pi}{5}n\right) + 2\cos\left(\frac{4\pi}{5}n\right) + 3\cos\left(\frac{6\pi}{5}n\right) \\ B = \text{Im}\{x[n]\} &= \sin\left(\frac{2\pi}{5}n\right) - 2\sin\left(\frac{4\pi}{5}n\right) + 3\sin\left(\frac{6\pi}{5}n\right) \end{aligned}$$

- (c) Let $a[n] = e^{i\frac{2\pi}{5}n}$, $b[n] = e^{-i\frac{4\pi}{5}n}$, and $c[n] = e^{i\frac{6\pi}{5}n}$. Are the signals a , b , and c linearly independent over the set of integers n ?

Answer: These signals are discrete-time complex exponentials, which can be interpreted as vectors (recall vector view of signals) in a function space (i.e., as sequences).

To check for linear dependence, we ask: Do there exist constants $\beta_1, \beta_2, \beta_3$, not all zero, such that:

$$\beta_1 e^{i\frac{2\pi}{5}n} + \beta_2 e^{-i\frac{4\pi}{5}n} + \beta_3 e^{i\frac{6\pi}{5}n} = 0 \quad \text{for all } n \in \mathbb{Z}?$$

Note that:

$$e^{-i\frac{4\pi}{5}n} = e^{i(-\frac{4\pi}{5}n)} = e^{i(\frac{6\pi}{5}n)} \quad \left(\text{since } -\frac{4\pi}{5} \equiv \frac{6\pi}{5} \pmod{2\pi}\right)$$

Another way to think about this equality can be found in the solution for part (a)

Therefore, $e^{-i\frac{4\pi}{5}n}$ and $e^{i\frac{6\pi}{5}n}$ are actually the same signal (up to periodic equivalence)! This means that among the three signals, two are the same (i.e., one is a linear combination of the other), implying linear dependence. So, the signals are NOT linearly independent.

Additionally, since the set $\{e^{i\frac{2\pi}{5}n}, e^{-i\frac{4\pi}{5}n}, e^{i\frac{6\pi}{5}n}\}$ has only two linearly independent elements, the span has dimension 2. Nice!